Cloud computing service models are best understood through the lens set out by Mell and Timothy (2011), who define the three main layers as Infrastructure as a Service, Platform as a Service and Software as a Service. AWS and Google Cloud both operate across these layers, but they approach flexibility and pricing quite differently, which shapes who tends to use each one.

AWS has been in the market longer and offers an enormous breadth of services, which gives it strong flexibility for organisations that want granular control over their infrastructure. Its pricing model relies heavily on pay as you go rates, reserved instances and spot pricing, which can get complex fast, but it rewards businesses that are willing to plan capacity carefully. Google Cloud, by contrast, tends to price things more simply and automatically applies sustained use discounts without requiring upfront commitments. This makes it more approachable for smaller teams or those without dedicated cloud finance expertise.

In terms of use cases, AWS is often the default choice for large enterprises running complex, hybrid or highly customised workloads, since its ecosystem and third party integrations are so extensive. Google Cloud tends to shine in data analytics and machine learning contexts, largely because of tools like BigQuery and its deep integration with TensorFlow, reflecting Google's own strengths in that space. Gupta (2024) makes a similar point, noting that provider choice is rarely just about cost but about which ecosystem aligns with an organisation's technical direction.

Ultimately, neither provider is universally better. The right choice depends on existing infrastructure, in house skills and long term technical strategy.

References

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Mell, P. and Timothy, G. (2011) The NIST Definition of Cloud Computing. Gaithersburg: U.S. Department of Commerce.